

# Learning with AI

## I2C Communication with Sensors

**Project:** Mold Growth Risk Predictor

**Name:** Idalia Martin

### Why I2C is Important

- Used to connect sensors in IoT systems
- Reduces wiring complexity
- Enables reliable communication between ESP32 and AHT10 sensor

Without I2C, the system cannot collect environmental data

## What I Learned

- How ESP32 communicates with AHT10 using SDA and SCL
- How to wire the sensor correctly (3.3V, GND, SDA, SCL)
- How to scan for I2C devices (address 0x38)
- How to read temperature and humidity data

## How I Applied It

- Connected AHT10 sensor to ESP32
- Used I2C scanner to detect the device
- Retrieved real-time temperature and humidity readings
- Debugged wiring and connection issues

## Challenges & Solutions

- Sensor not detected → fixed wiring
- Wrong pin mapping → learned GPIO vs labeled pins
- Library errors → installed correct Adafruit library

## How AI Helped Me

- Explained I2C communication in simple terms
- Helped debug connection issues
- Interpreted error messages
- Guided correct wiring and setup

# Reflection

- I started with no experience in hardware communication
- Learned how sensors communicate with microcontrollers
- Successfully built a working sensor system